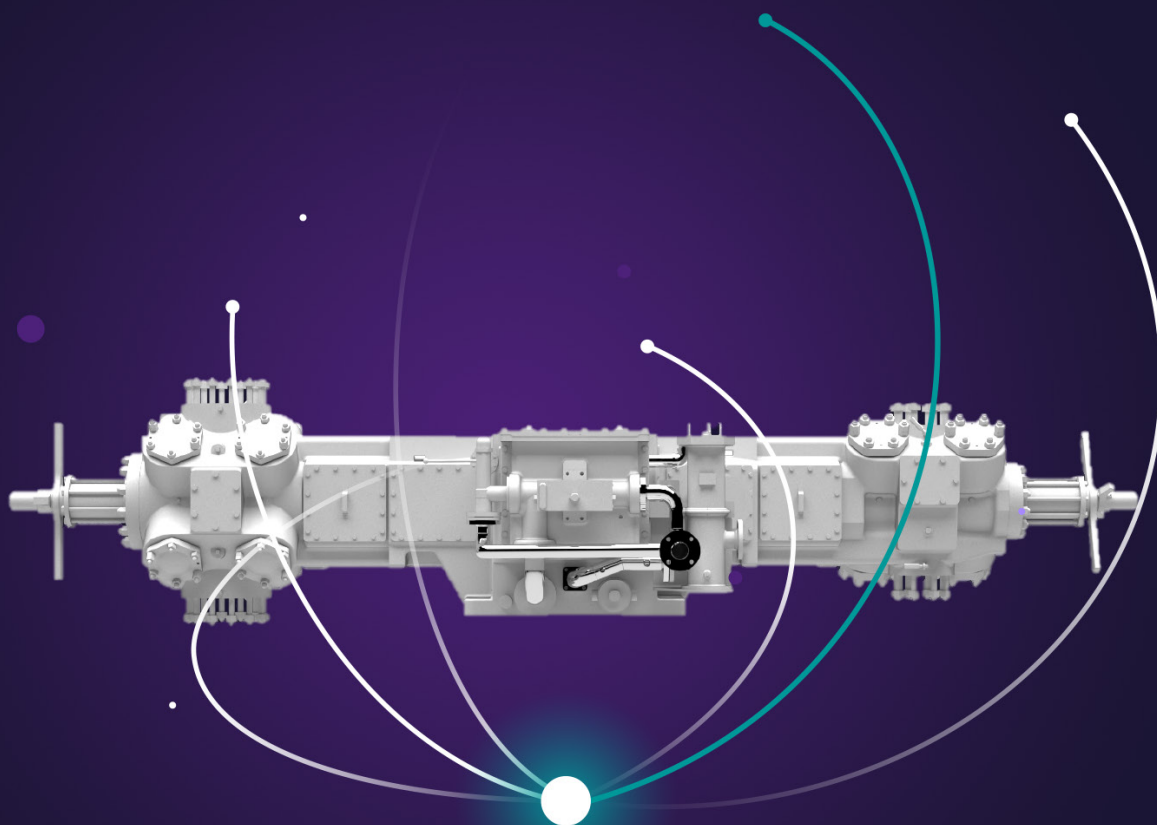


# HOSS technical datasheet



## General information

|                                                     |             |             |             |
|-----------------------------------------------------|-------------|-------------|-------------|
| Max allowable continuous combined rod load (T or C) | 75,000 lbs  |             |             |
| Max allowable continuous gas load (T or C)          | 87,000 lbs  |             |             |
| Total gas load                                      | 174,000 lbs |             |             |
| Piston rod diameter                                 | 2.875"      |             |             |
| Stroke                                              | 5.00"       | 6.00"       | 7.00"       |
| Maximum rated speed                                 | 1500 RPM    | 1200 RPM    | 1000 RPM    |
| Minimum rated speed                                 | 500 RPM     | 500 RPM     | 500 RPM     |
| Average piston speed at max RPM                     | 1250 ft/min | 1200 ft/min | 1167 ft/min |

## Frame information

|                                                          |                            |            |             |
|----------------------------------------------------------|----------------------------|------------|-------------|
| Number of throwsw                                        | 2 throw                    | 4 throw    | 6 throw     |
| Crankcase oil capacity                                   | 34 gallons                 | 69 gallons | 105 gallons |
| Oil pump capacity at 1000 rpm                            | 63 GPM                     | 94 GPM     | 125 GPM     |
| Oil flow to cooler/filter                                | 32 GPM                     | 64 GPM     | 96 GPM      |
| Normal frame oil temperature                             | 150°F - 170°F              |            |             |
| Maximum frame oil temperature                            | 180°F                      |            |             |
| Normal oil header pressure                               | 50 PSI - 65 PSI            |            |             |
| Minimum oil header pressure                              | 40 PSI                     |            |             |
| Oil header pressure shutdown setting                     | 35 PSI                     |            |             |
| Minimum start-up oil temperature                         | 80°F                       |            |             |
| Minimum frame oil temperature to load                    | 90°F                       |            |             |
| Maximum oil filter differential pressure                 | 15 PSI                     |            |             |
| Crankshaft thrust                                        | .014" - .027"              |            |             |
| Frame mounted oil pump thrust                            | .006" - .009"              |            |             |
| Oil pump coupling hub offset                             | .250"                      |            |             |
| Vertical frame thermal growth                            | .007" - .010"              |            |             |
| Balance tolerance of opposing throws - onshore           | +/- 2 lbs                  |            |             |
| Balance tolerance of opposing throws - offshore          | +/- 1 lbs                  |            |             |
| Maximum distance piece internal pressure                 | 25 PSI                     |            |             |
| Top main bearing clearance                               | .005" - .010" feeler gauge |            |             |
| Connecting rod bearing to crankpin diametrical clearance | .006" - .011" micrometer   |            |             |
| Crosshead pin to crosshead diametrical clearance         | .001" - .003" micrometer   |            |             |
| Crosshead pin to conn rod bushing diametrical clearance  | .006" - .011" micrometer   |            |             |
| Cold top crosshead shoe to guide clearance               | .013" - .018" feeler gauge |            |             |

## Cylinder information

|                                              |                                 |                                  |                                  |                                  |                                  |                                  |                         |               |
|----------------------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|-------------------------|---------------|
| Frame end cold bump clearance                |                                 |                                  |                                  |                                  |                                  |                                  |                         | .043" - .082" |
| Outer end cold bump clearance                |                                 |                                  |                                  |                                  |                                  |                                  |                         | .064" - .123" |
| Maximum gas discharge temperature            |                                 |                                  |                                  |                                  |                                  |                                  |                         | 350 °F        |
| Cooled packing coolant flow rate per packing |                                 |                                  |                                  |                                  |                                  |                                  |                         | 2.9 GPM       |
| Normal coolant pressure                      |                                 |                                  |                                  |                                  |                                  |                                  |                         | 35 PSI        |
| Maximum coolant pressure                     |                                 |                                  |                                  |                                  |                                  |                                  |                         | 75 PSI        |
| Maximum cold horizontal piston rod runout    |                                 |                                  |                                  |                                  |                                  |                                  |                         | .003"         |
| Maximum cold vertical piston rod runout      |                                 |                                  |                                  |                                  |                                  |                                  |                         |               |
| 6.00" - 7.75"<br>cylinder dia.               | 8.00" - 10.50"<br>cylinder dia. | 11.50" - 13.00"<br>cylinder dia. | 14.00" - 16.25"<br>cylinder dia. | 17.50" - 19.00"<br>cylinder dia. | 20.50" - 23.00"<br>cylinder dia. | 24.50" - 26.50"<br>cylinder dia. | 28.00"<br>cylinder dia. |               |
| .0000" - .0020"                              | .0005" - .0025"                 | .0015" - .0035"                  | .0025" - .0045"                  | .0030" - .0055"                  | .0040" - .0060"                  | .0045" - .0065"                  | .0050" - .0070"         |               |

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